

[←](#) Back to the profile of Ankit Bhardwaj (/profile?id=~Ankit_Bhardwaj1)

FieldFormer: Locality-Aware Transformers for Spatio-Temporal Modeling on Sparse Sensor Networks



Ankit Bhardwaj (/profile?id=~Ankit_Bhardwaj1),
Ananth Balashankar (/profile?id=~Ananth_Balashankar1),
Lakshmi Subramanian (/profile?id=~Lakshmi_Subramanian1) 

 18 May 2026 (modified: 26 May 2026)

 Under review for TMLR

 Everyone

 Revisions (/revisions?id=we4FYGOE2y)

 BibTeX

 CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>)

Edit ▾

Abstract: Spatio-temporal sensor data in real-world systems is often sparse, noisy, and irregular, making it difficult to infer global structure from limited observations. Under extreme sparsity, we run into the limits of identifiability of latent system states, making latent field reconstruction fundamentally underconstrained. In such scenarios, multiple physically plausible fields may remain consistent with the same observations, requiring reconstruction models to rely heavily on inductive biases regarding locality, transport structure, and spatial regularity.

Under such sparsity regimes, reliable reconstruction becomes concentrated around the observational support induced by the sensor network, making sensor-space modeling a more identifiable objective than unconstrained global field recovery. We introduce FieldFormer, a mesh-free transformer architecture designed for locality-aware sensor-space modeling in persistent sensor networks. For each query, FieldFormer aggregates local evidence using a learnable velocity-scaled distance metric that adapts neighborhood geometry to heterogeneous spatio-temporal relationships. Neighborhoods are constructed as fixed maximal sparse contexts over nearby sensors and bounded temporal windows, while learned velocity-scaled offsets modulate token geometry within this context, enabling stable and scalable inference under extreme sparsity. A local transformer encoder integrates neighborhood information, while global consistency is modeled through coordinate-based neural field formulation.

We evaluate FieldFormer across five benchmarks spanning synthetic and real-world spatio-temporal systems, including anisotropic heat diffusion, shallow-water dynamics, atmospheric transport fields, and pollution monitoring datasets. Our results reveal that locality-aware reconstruction provides strong advantages in persistent sparse sensor networks where local domains of dependence remain observed, enabling FieldFormer to consistently outperform state-of-the-art baselines on sensor-space prediction tasks under highly sparse and noisy sensing regimes.

Submission Type: Long submission (more than 12 pages of main content)

Competing Interests:  No competing interests.

Human Subjects Reporting:  N/A

Assigned Action Editor: Taylor W. Killian (/profile?id=~Taylor_W_Killian1)

Submission Number: 9033

Filter by reply type... ▾

Filter by author... ▾

Search keywords...

Sort: Newest First



Everyone
 Editors In Chief
 Paper9033 Action...
 Paper9033 Authors

5 / 5 replies shown

Add:

Withdrawal

Official Comment

Its time for the discussion period!

Official Comment by Action Editor z24K (Taylor W. Killian (/profile?id=~Taylor_W_Killian1))

29 Jun 2026, 14:11 Everyone

Comment:

Hi everyone!

We now have all three reviews for this paper. First, I wanted to briefly thank the reviewers for getting them in on time and with great quality. There is a lot of useful feedback for the authors.

I want to remind you, as per TMLRs review procedure, that we are given two weeks for the discussion phase. This includes iterations of author responses and clarifications, possible revisions to the paper (including experiments), and reviewer feedback. For this to be a successful discussion period, I invite both authors and reviewers to be engaged and generous.

As a note to all reviewers, I will not consider your final recommendations unless I see good faith efforts to engage with the authors during this discussion period. We owe them and their work this respect.

Best, Taylor

Add:

Official Comment

Review of Paper9033 by Reviewer hGFy

Review by Reviewer hGFy 27 Jun 2026, 02:31 (modified: 27 Jun 2026, 02:31) Everyone

Revisions (/revisions?id=vchjPYrJzc)

Summary Of Contributions:

This paper studies a spatio-temporal field reconstruction problem, and the challenge is learning from sparse sensor dataset to predict the field at unobserved locations. The authors propose a new architecture called FieldFormer, which is a locality-aware transformer designed for sparse sensor networks. In this new algorithm, the learner gathers a fixed-size candidate set of nearby sensor-time tuples for a query point, encodes each neighbor by its relative space-time offset and observed value, scales those offsets by a learnable velocity-scaled (anisotropic) metric, passes the resulting tokens through a small transformer encoder, mean-pools, and maps to the predicted field value with an MLP head. The authors also provide two variants of FieldFormer: FieldFormer-PINN and FieldFormer--Gamma Field. The authors provide two theoretical results, one is about the expressivity of FieldFormer, and the other is about the miss-coverage lower bound. The authors also conduct extensive experiments on three synthetic PDEs and two real-world datasets, and compare FieldFormer with several baselines.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: Yes

Explain your answer above:

The author gives a strong theoretical guarantee is Theorem 5.1 to support the expressivity of FieldFormer, and Theorem 5.3 to support the miss-coverage lower bound. The author also provides extensive experiments to support the effectiveness of FieldFormer. However, there are some issues with the theoretical results and the empirical results.

1. For the expressivity result, I notice that the proof of Theorem 5.1 only shows that FieldFormer can approximate the one-step explicit finite-difference update map ψ to arbitrary accuracy when the full, exact stencil is supplied. However, it does not show that FieldFormer can uniformly approximate the true continuous solution u . Therefore, the expressivity result is only partially supported.

2. For the miss-coverage lower bound, I did not follow the step 3 of the proof, where the author sums over unobserved spatial sites and their time levels. I think this step is not valid, because the conditional risk R is a single nonnegative scalar, and many lower bounds on a single nonnegative scalar combine by max, not sum. Could the author clarify this step?
3. For the empirical results show that FieldFormer outperforms the baselines on the real-world sensor-space imputation tasks, but it does not consistently outperform the baselines on the synthetic sensor-space tasks. Therefore, the claims made in the submission are only partially supported by accurate, convincing and clear evidence.

Would at least some individuals in TMLR's audience be interested in knowing the findings of this paper?: Yes
Explain your answer above:

Yes. Spatio-temporal reconstruction from sparse, persistent sensor networks is of broad interest across the TMLR community.

Requested Changes:

Please see the above discussion on the partial support of the theoretical claims, the overstatement on the empirical claims.

Broader Impact Concerns:

No broader impact statement is needed.

Add: **Official Comment**

Review of Paper9033 by Reviewer NuTG

Review by Reviewer NuTG 📅 22 Jun 2026, 07:53 (modified: 27 Jun 2026, 02:31) 👁 Everyone

📄 Revisions (/revisions?id=Klw5ZH2bFZ)

Summary Of Contributions:

This paper studies spatiotemporal physical field modeling in sparse, noisy, irregular, and long-term deployed sensor networks. Its central position is that, under extreme spatial sparsity, recovering a complete global latent field from only a small number of sensor trajectories is generally non-identifiable. Therefore, a more appropriate objective is sensor-space imputation on the persistent sensor support, while interpreting predictions far from the observational support as prior-guided surrogate fields. Methodologically, the paper proposes FieldFormer: for each query, it constructs a fixed-size maximal local spatiotemporal context, rescales relative coordinate offsets using a velocity-scaled learnable metric, and aggregates neighborhood tokens with a local Transformer encoder. The paper also provides a local expressivity theorem and a miss-coverage lower bound to explain the relationship between local observational coverage and recoverability.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: Yes

Explain your answer above:

The paper provides substantial empirical evidence through experiments on synthetic and real-world datasets, comparisons with multiple baselines, and ablation studies. While some claims could be stated more cautiously, the main conclusions are generally supported by the presented results.

Would at least some individuals in TMLR's audience be interested in knowing the findings of this paper?: Yes

Explain your answer above:

The paper studies an important problem in spatio-temporal modeling and sparse sensing. Its perspective on sensor-space imputation under extreme spatial sparsity is likely to be of interest to part of the TMLR audience.

Requested Changes:

Weaknesses

1. **The current theoretical proofs are insufficient to support the level of theoretical credibility that the paper attempts to establish.** My specific concerns are listed under "Questions." The authors should carefully check whether there are errors in the formula derivations and whether there are questionable logical gaps throughout the paper.

2. The proposed FieldFormer relies on local observational support and has structural limitations for unseen-sensor prediction and global reconstruction. The paper's view that "under extreme spatial sparsity, global field reconstruction is intrinsically non-identifiable, and therefore the focus should be on sensor-space imputation" is reasonable and insightful. However, this also means that FieldFormer's advantage depends heavily on the existence of sufficient observational support near the query location. When an entire site is held out, or when predictions are required in new regions far from existing sensors, the local evidence available to FieldFormer is substantially reduced. In such cases, methods with stronger global priors may perform better. Table 4 also shows that, in the real sensor-holdout setting, ImputeFormer / Senseiver outperform FieldFormer on atmospheric AT/RH and pollution PM10/PM2.5.

Questions

1. Theorem 5.1 seems to claim that a continuous PDE solution $u(z)$ can be uniformly approximated on a compact domain, namely

$$\sup_{z \in K} \|u(z) - \hat{u}(z)\| < \varepsilon.$$

However, from the proof, what is actually established seems to be only that FieldFormer can approximate the discrete update map of a finite stencil,

$$u^{n+1}(i) = \psi(u^{n-\ell}(i+s) : (s, \ell) \in S),$$

that is,

$$\sup_{v \in V_K} \|\psi(v) - \hat{\psi}(v)\| < \varepsilon.$$

The latter does not directly imply the former. The transition from approximating a finite-stencil update rule to uniformly approximating the solution of a continuous PDE requires a standard numerical-analysis convergence argument, usually involving an error decomposition such as

$$\|u - \hat{u}\| \leq \underbrace{\|u - u_h\|}_{\text{discretization error}} + \underbrace{\|u_h - \hat{u}\|}_{\text{network approximation error}},$$

where u_h denotes the numerical solution produced by a finite-difference scheme. From the current proof, it seems that only the second term is controlled, while the first term $\|u - u_h\|$ is not.

Could the authors further clarify what additional assumptions are needed to control $\|u - u_h\|$? If the main purpose of this result is actually to show that FieldFormer has sufficient expressivity to approximate finite-stencil dynamics, would it be more accurate to state the theorem directly as an approximation result for the update map ψ , rather than as a uniform approximation result for the continuous PDE solution $u(z)$?

2. Regarding Theorem 5.3, in Eq. (2) of Appendix C, the authors obtain the inequality $\frac{1}{2}(a+b) \geq \frac{D}{2}$, where $a = \mathbb{E}[\|\hat{u}(V) - \psi(V)\|_2 \mid I]$, $b = \mathbb{E}[\|\hat{u}(TV) - \psi(TV)\|_2 \mid I]$, and $D = \mathbb{E}[\|\psi(V) - \psi(TV)\|_2 \mid I]$. However, this only implies $\max a, b \geq D/2$, and does not directly imply $a \geq D/2$.

In addition, in the subsequent proof, the authors assume

$$\mathbb{E}[\|\psi(V) - \psi(T_{(s,\ell)}V)\|_2] \geq ca(s, \ell).$$

But this inequality cannot simply be summed over all s, ℓ , because the effects of different coordinates may interact with each other and may even partially cancel out. Therefore, stronger assumptions may be needed.

3. Regarding the learned metric, Eq. (2) of the paper uses an axis-aligned diagonal metric and adjusts the relative scales of the spatial and temporal dimensions through learnable parameters $\gamma_x, \gamma_y, \gamma_t$. This design helps learn the trade-off between spatial and temporal scales, but it cannot directly characterize characteristic-aligned

transport, where the relevant upstream observations satisfy a relation such as $\Delta x \approx v \Delta t$.

Have the authors considered using a static flow-aligned metric of the following form?

$$d(i) = (\Delta x_i - v \Delta t_i)^\top M (\Delta x_i - v \Delta t_i) + \gamma_i^2 \Delta t_i^2,$$

where v and $M \succeq 0$ are global learnable parameters. This design could express characteristic-aligned transport and rotational anisotropy.

Broader Impact Concerns:

I do not have major broader-impact concerns.

Add: [Official Comment](#)

Review of Paper9033 by Reviewer jzvM

Review by Reviewer jzvM  19 Jun 2026, 17:47 (modified: 27 Jun 2026, 02:31)  Everyone

 Revisions (/revisions?id=fVyoTODKUB)

Summary Of Contributions:

- The paper studies spatio-temporal reconstruction from persistent sparse sensor networks—settings in which observations are dense in time but limited to a small fixed set of spatial locations.
- It introduces a conceptual distinction between sensor-space modeling (predicting at observed sensor locations) and unconstrained global field reconstruction, and argues that the former is the more identifiable objective under extreme sparsity.
- It proposes FieldFormer, a mesh-free architecture that constructs an adaptive local neighborhood around each query via a learnable velocity-scaled distance metric, encodes relative offsets and observations as tokens, and aggregates them with a local transformer encoder. Two variants are also introduced: FieldFormer-PINN (with differentiable PDE-residual regularization) and FieldFormer-Gamma Field (with a coordinate-dependent metric).
- Two theoretical results are presented: Theorem 5.1 (expressivity for locally structured spatio temporal dynamics) and Theorem 5.3 (average-case lower bound under longitudinal sampling).
- Empirically, FieldFormer is compared against eight baselines across three synthetic PDE benchmarks and two real-world datasets, evaluated under three regimes: sensor-space imputation, sensor-holdout generalization, and global field reconstruction. Ablations isolate the contributions of the PINN variant, the Gamma Field variant, and the local transformer (vs an MLP variant).
- An anonymized code repository is provided for review.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: No

Explain your answer above:

Several core claims are not adequately supported by the evidence presented. These issues are recoverable through revision.

1. **False dichotomy in the conceptual framing.** The paper contrasts sensor-space modeling against unconstrained global reconstruction and concludes the former is the right objective. This ignores the standard alternative for under-determined problems: methods that genuinely retain multiple plausible solutions (e.g., CSDI, ConvCNP/ConvNP, Neural Processes, deep ensembles evaluated by predictive distribution). The only baseline labeled probabilistic (SVGP) is, in practice, a variationally-optimized single-solution method evaluated by its posterior mean—it does not represent the class of uncertainty-expressive methods that the paper’s identifiability argument calls for. The paper’s “we included a probabilistic baseline” framing therefore rests on a category mismatch.
2. **Local decomposition framed as a new paradigm.** Operationally FieldFormer performs per-query local prediction; the repeated “globally grounded through continuous coordinates” framing is not substantiated by any ablation. No baseline isolates the contribution of the local-decomposition strategy from that of the specific architecture, so the contribution of the decomposition strategy cannot be separated from that of the specific

architecture. Table 5 already shows FieldFormer-MLP within 0.3% of full FieldFormer on Heat, reinforcing this concern.

3. **Reporting does not match stated conventions.** Captions for Tables 1, 2, 4–8 state “mean $\pm$ standard deviation over bootstrap samples,” but the tables contain only point estimates. Comparisons differing by < 0.01 (e.g., Table 1 Heat RMSE) cannot be assessed for statistical significance.
4. **Experimental fairness requires more transparency.** The mesh-dependent baselines (RecFNO, ImputeFormer, Senseiver) are adapted to the sparse setting via rasterization, fixed-node windows, or global-latent bottlenecks, which may push them outside their design regimes. An explicit table comparing each method’s loss function, input representation, and hyperparameter search budget would clarify whether the comparison is like-for-like.

Would at least some individuals in TMLR’s audience be interested in knowing the findings of this paper?: Yes
Explain your answer above:

- Sparse spatio-temporal reconstruction is a substantive problem for TMLR’s audience, with active research communities in spatial statistics, geostatistics, environmental ML, physics-informed learning, and operator learning. The mesh-free design and the explicit attention to evolving sensor topologies have practical appeal beyond the specific datasets used.
- Even with the concerns above, the conceptual question the paper raises—how to think about reconstruction quality when the observation geometry itself is the bottleneck—is one the community should engage with. A revised version that addresses the comparison gaps and reporting issues would be of clear interest to readers working on PDE learning, neural fields, and applied sensor data.

Requested Changes:

Critical

- Add genuinely uncertainty-expressive baselines. Include at least one baseline that genuinely retains and is evaluated on its predictive distribution. SVGP under variational approximation is effectively a single-solution method and does not represent this class.
- Add ablation baselines that isolate the contribution of local decomposition from the architecture. The current results do not separate (i) the effect of restricting prediction to a local neighborhood from (ii) the effect of the specific transformer-based aggregation within that neighborhood. Table 5 (FieldFormer-MLP within 0.3% of full FieldFormer on Heat) already suggests this separation is needed. Suitable baselines are at the authors’ discretion.
- Report bootstrap standard deviations on Tables 1–8, as the captions promise; tables currently show only point estimates.
- Make the like-for-like nature of the baseline comparison verifiable. Readers should be able to confirm that the compared methods are solving comparable problems with comparable tuning effort. The form of disclosure (e.g., a protocol table, expanded text, or appendix) is at the authors’ discretion.

Important

- Substantiate scalability and efficiency claims empirically. The current claims about mesh-free efficiency rest on qualitative statements; concrete empirical comparison against the baselines is needed. Which efficiency metrics best capture the claim is at the authors’ discretion.
- Strengthen the Section 1 motivation beyond ML-internal references. The real-world stakes of persistent sparse sensor networks are currently asserted but supported only by closely-related ML papers. Citations from the application domains themselves (e.g., environmental monitoring, observation system design, IoT infrastructure) would establish the broader stakes the framing relies on. Selection of appropriate sources is at the authors’ discretion.
- Address diffusion-based imputation (CSDI and related score-based methods); include as baselines or justify the omission in Related Work.

Minor

- Define paper-specific terminology at first use (“velocity-scaled distance metric,” “fixed maximal sparse context,” “token geometry”), or replace with standard terms. The Abstract and the Section 1 paragraph beginning “For each query location...” propagate this issue.
- Improve self-containedness around the Transformer encoder. The paper’s novelty claim is that the learned metric modulates attention, which is hard for readers outside core ML to evaluate without an explicit treatment of the

attention machinery being modified. The level and placement of background material is at the authors' discretion.

- Substantiate or remove unsupported framings. The “globally grounded through continuous coordinates” phrase is never shown to be load-bearing; either support it or rephrase. The “naturally adapts” claim in Section 4.1 is similarly qualitative and would benefit from quantitative grounding.

Broader Impact Concerns:

The paper includes a Broader Impact section that appropriately notes (i) the wide deployment of sparse sensor networks in environmental, infrastructure, and scientific contexts, (ii) that global reconstruction under extreme sparsity is generally not identifiable, and (iii) that outputs should be interpreted as prior-guided estimates rather than ground truth far from sensor support. This framing is reasonable and largely sufficient.

Add:

Official Comment

Review Approval of Paper9033 by Action Editor z24K

Review Approval by Action Editor z24K (🌐 Taylor W. Killian (/profile?id=~Taylor_W_Killian1))
📅 26 May 2026, 14:04 (modified: 26 May 2026, 14:04) 👁 Editors In Chief, Action Editors, Authors
📄 Revisions (/revisions?id=BRjTWHrnTL)

Under Review: Appropriate for Review

Add:

Official Comment

Contact (/contact)

- About OpenReview (/about)
- FAQ (https://docs.openreview.net/getting-started/frequently-asked-questions)
- Hosting a Venue (/group?id=OpenReview.net/Support)
- Sponsors (/sponsors)
- Terms of Use (/legal/terms) / Privacy Policy (/legal/privacy)
- All Venues (/venues)
- Donate** (/donate)

OpenReview (/about) is a long-term project to advance science through improved peer review with legal nonprofit status. We gratefully acknowledge the support of the OpenReview Sponsors (/sponsors). © 2026 OpenReview